AUTOMATED DETECTION OF SATELLITE TRAILS IN GROUND-BASED OBSERVATIONS

U-Net segmentation with probabilistic Hough post-processing

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Motivation & Scientific Justification

Low-Earth-orbit satellite constellations increasingly cross wide-field astronomical exposures, leaving bright linear trails that corrupt photometry, astrometry, and transient-source catalogues. Manual inspection does not scale to survey operation, so observatories need automated trail masks that are reliable at the pixel level and cheap enough to run inside reduction pipelines.

This project replicates the satellite-trail detector of Stoppa et al. (2024): a U-Net trained on labelled MeerLICHT image patches, followed by a probabilistic Hough transform that reconnects linear trail evidence missed by the network. The replication is used as a diagnostic tool: the aim is not only to reproduce a headline score, but to identify where the residual error comes from.

Key Findings

The replication reproduces the target paper's recall closely: five-seed sampled-test recall is 0.923 ± 0.007 against the paper's reported 0.94. Strict precision is lower, at 0.847 ± 0.010 on the sampled test set and 0.780 on the parity set that restores the natural negative-patch prevalence. This shows that precision is prevalence-sensitive, whereas recall is the cleaner like-for-like comparison across the two test populations. Image-cluster uncertainty is the relevant interval because source images, not patches, are the independent units.

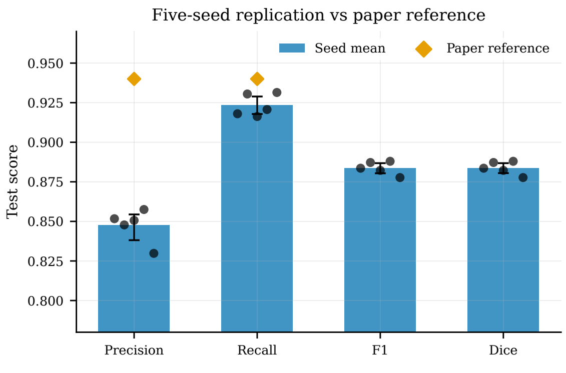

Figure 1: Five-seed replication performance for the locked U-Net compared with the Stoppa et al. reference. Paper-reference markers are shown only for the explicitly reported precision and recall values; recall is closely reproduced while strict precision remains lower, motivating the boundary-scale error analysis.

The Hough stage reproduces the paper's intended completeness behaviour. On the full-coverage tiled canvas, trail-pixel recall rises from 0.918 to 0.990, equivalently reducing the pixel-level false-negative rate from 0.082 to 0.0105. This does not make the Hough mask a better strict pixel classifier: post-Hough line drawing increases false-positive area, reducing strict precision from 0.780 to 0.410 on the parity canvas.

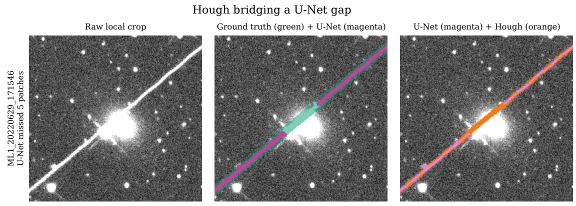

Figure 2: Hough bridging a U-Net gap over a bright star. The example illustrates the mechanism: linear post-processing recovers ground-truth trail pixels missed by the thresholded U-Net.

The strict precision gap is localised to boundary scale. False-positive decomposition shows that only 17.5 % of false-positive pixels occur in trail-free patches; the remaining 82.5 % fall within trail-bearing patches. A one-pixel boundary tolerance raises five-seed precision from 0.847 to 0.946, and 57 % of all false positives lie within one pixel of a labelled trail. These diagnostics are consistent with the residual error being concentrated near labelled trail boundaries. A blinded single-author re-annotation of 64 sealed crops is consistent with this: the original masks, the re-annotation, and the model all share a median width of 6 px, and the model agrees with the re-annotation about as well as the original labels do, with no strict-F_1 difference resolved under this design (paired -0.0001, 95% CI spanning zero). This is a single-author between-source disagreement estimate, not inter-annotator agreement. A small over-firing residual remains on adversarial controls, so the audit is consistent with boundary-scale disagreement rather than a uniformly thin label or model over-paint.

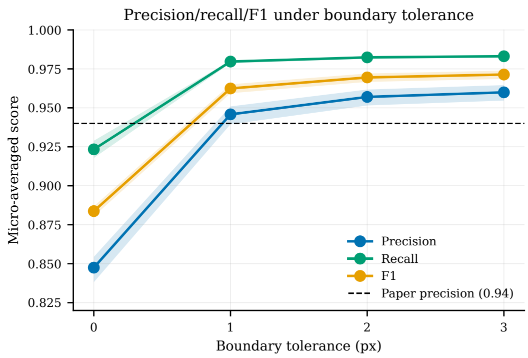

Figure 3: A one-pixel tolerance largely closes the strict precision gap, consistent with residual disagreement concentrated near labelled boundaries.

Methodology

The study uses 178 labelled MeerLICHT image/mask pairs split at image level into 122 training, 24 validation, and 32 test images, avoiding same-image leakage. Images are tiled into 528×528 patches. The U-Net is reimplemented in PyTorch from the released Keras model configuration, preserving average pooling, LeakyReLU activations, in-block dropout, Keras-style initialisation, and a combined binary cross-entropy plus squared-Dice loss.

Hyperparameters are selected by validation-only Optuna search, the top trials are retrained across five seeds, and the final threshold is chosen from a validation precision–recall sweep before test metrics are inspected. Hough post-processing is evaluated on a full-coverage tiled canvas that reproduces the parity canvas pixel-for-pixel, with separate precision and completeness caveats.

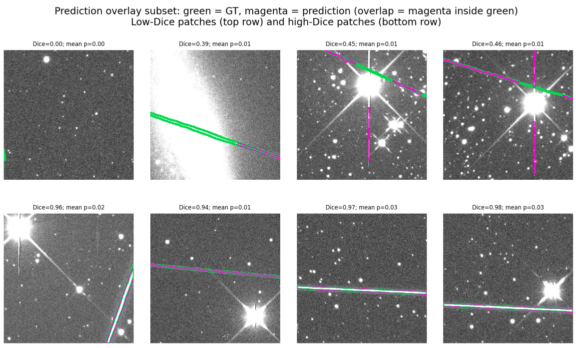

Figure 4: Diagnostic selection of test-patch predictions for the locked U-Net: ground truth in green, prediction in magenta, and overlap inside the green band.

Extensions & Transfer

Follow-up experiments test whether the residual can be moved by model or inference changes. A classifier-gated detector shifts precision by only +1.2 percentage points, consistent with the 17.5 % inter-patch upper bound. An attention-gated U-Net at matched capacity is a null result, clDice shows no hidden topology gain, hard-negative mining adds only +0.007 precision, and ensembling with test-time augmentation sits within seed noise.

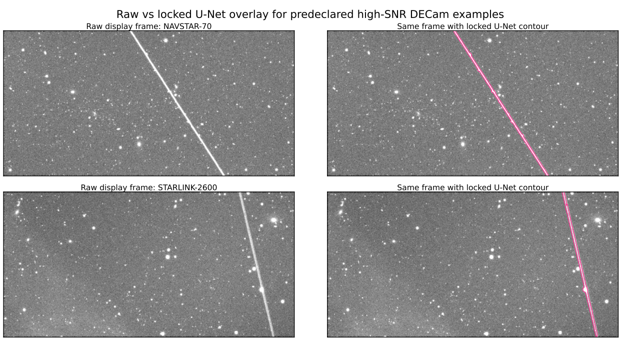

Figure 5: Cold-domain DECam examples: raw measured-streak frames and the locked MeerLICHT-trained U-Net contour. This is a qualitative transfer illustration, not a benchmark.

Recommendation & Impact

The practical result is a high-completeness trail masker whose remaining strict-precision gap is mostly boundary-near. The appropriate reporting standard is therefore strict precision/recall plus a boundary-tolerant operating point and an explicit mask-width convention. The strongest remaining check is a multi-annotator annotation audit, which would establish inter-annotator agreement and test how much of the strict precision gap reflects mask granularity rather than detector error. A labelled cross-instrument test, especially on DECam-like data, is the next generalisation check; the current DECam application is qualitative only.

The U-Net plus Hough pipeline closely reproduces the target detector's recall and Hough completeness. The tested capacity, training-protocol and post-processing changes do not materially reduce the remaining strict-precision residual. For operational masking, the evidence points to reporting strict scores alongside boundary-tolerant scores and annotation conventions rather than prioritising another architecture tweak.

References

[1] F. Stoppa et al., "Automated detection of satellite trails in ground-based observations using U-Net and Hough transform," *Astronomy & Astrophysics*, 692, A199, 2024. [2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," MICCAI, 2015. [3] J. Matas, C. Galambos, and J. Kittler, "Robust detection of lines using the progressive probabilistic Hough transform," *Computer Vision and Image Understanding*, 78(1), 2000.

Declaration of Use of Autogeneration Tools

I have used Codex (ChatGPT) and Claude Code to support me across this work, primarily in the following areas:

- Providing advice and usage support for libraries like PyTorch and PIL.
- Generating boilerplate code for plotting, testing and scripting.
- Generating function docstrings.
- Reviewing and auditing project structure for completeness, consistency, accuracy and best practices in software engineering.